

```
$ while true; do cat BigFile.txt > /dev/null;done &
```

```
$ echo
$ echo none
```

```
$ echo none > /sys/block/sda/queue/scheduler
```

```
$ time find ../../../../Android/Sdk/. -type f -exec cat '{}' ';' > /dev/null # path to Android Sdk
```

The test results obtained for various I/O schedulers are given in Table 5.

The performance evaluation of different I/O schedulers (*none*, *kyber*, *bfq*, and *mq-deadline*) was conducted, and the results showed that the *none* I/O scheduler outperformed the others in both Test 1 and Test 2. Interestingly, all the I/O schedulers performed better when used inside a container. However, the *none* I/O scheduler proved to be the most suitable choice for the given test conditions, as it provided the best results for both tests.

These findings are particularly relevant for processes that involve intensive disk access. In such cases, reading from a Docker container is faster and can significantly improve performance. For instance, if there are numerous reading and writing operations, such as reading a .csv file for data analytics and storing the results in a file, the *none* I/O scheduler can be particularly useful in enhancing performance. END 🐼

References

- Love, R. (2004, February 1). Kernel Korner - I/O Schedulers. Linux Journal. Retrieved January 14, 2023, from <https://www.linuxjournal.com/article/6931>
- Ubuntu Wiki (2019, September 10). Kernel/Reference/IOSchedulers. Ubuntu Wiki. Retrieved January 14, 2023, from <https://wiki.ubuntu.com/Kernel/Reference/IOSchedulers>

Table 4: kyber I/O scheduler results

File size	Real-time in Ubuntu 20.04 (in seconds)	Real-time in Docker and container (in seconds)	Percentage improvement
10MB	0.013	0.002	84.62
20MB	0.018	0.004	77.78
30MB	0.093	0.004	95.70
40MB	0.073	0.006	91.78
50MB	0.074	0.006	91.89
60MB	0.144	0.009	93.75
70MB	0.111	0.01	90.99
80MB	0.072	0.014	80.56
90MB	0.259	0.014	94.59
100MB	0.162	0.016	90.12

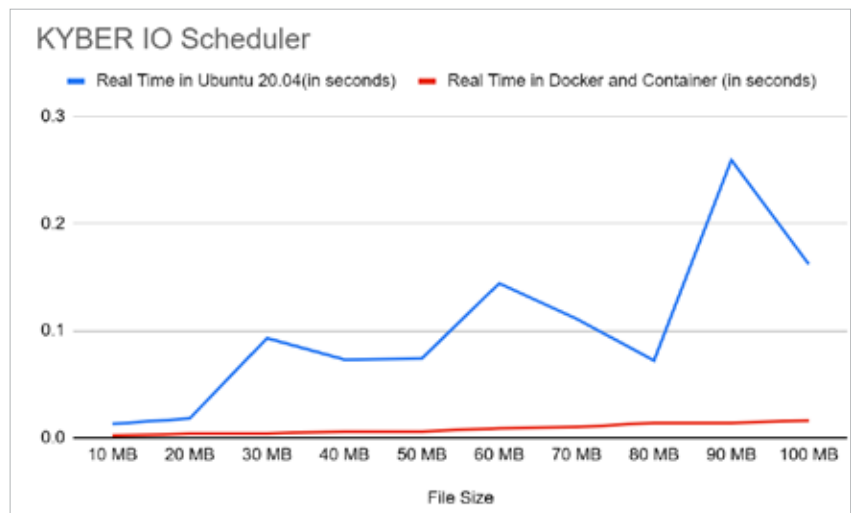


Figure 4: Graphical performance comparison of *kyber* I/O schedulers in VM vs Docker container

Table 5: Test results obtained for various I/O schedulers

I/O scheduler	VM (in seconds)	Container (in seconds)	Percentage improvement
none	94.133	79.805	17.9537623
mq-deadline	99.765	82.228	21.32728511
kyber	100.315	86.747	15.64088672
bfq	114.214	92.577	23.37189583

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